1.3: Rates of Change and Behavior of Graphs

Definition. (Rate of Change)
A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are "output units per input units". The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Example. Suppose the following table contains temperatures recorded through the day.

h	7 AM	9 AM	11 AM	1 PM	3 PM
$T(h)$	$78°F$	$81°F$	$87°F$	$93°F$	$98°F$

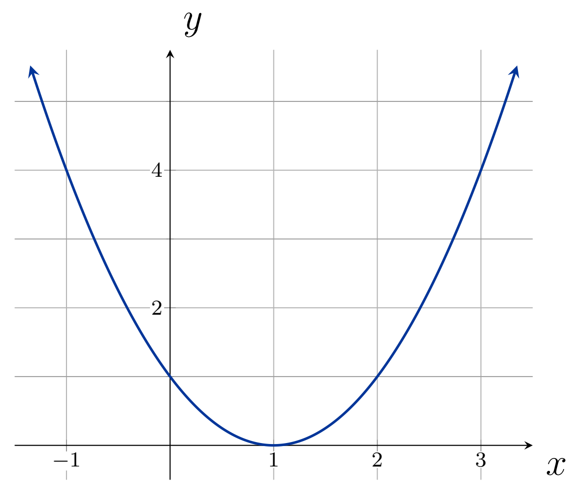
Find the average rate of change between

7 AM and 9 AM

9 AM and 11 AM

7 AM and 3 PM

Example. Using the following graph, determine the average rate of change on the interval $[-1, 2]$



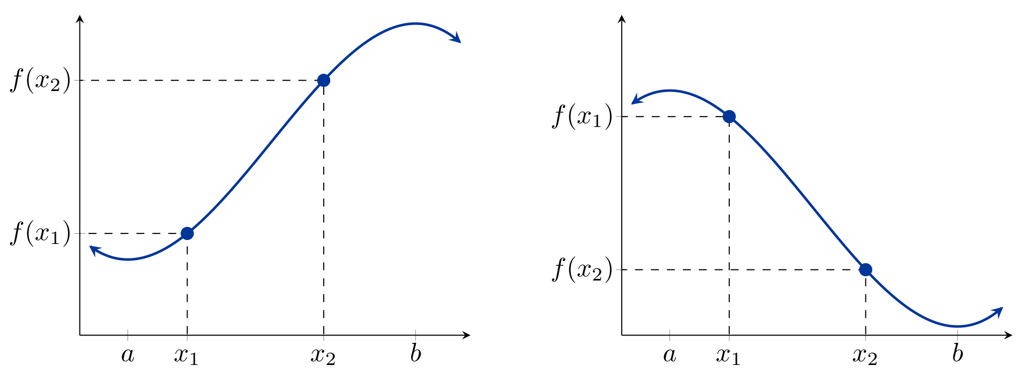
Example. Find the average rate of change of $f(x) = x^2 - \dfrac{1}{x}$ on the interval $[2, 4]$.

Example. Find the expression for the average rate of change of $g(t) = t^2 + 3t + 1$ on the interval $[0, a]$.

Example. Find the expression for the average rate of change of $j(t) = \dfrac{1}{t+3}$ on the interval $[1, 1+h]$.

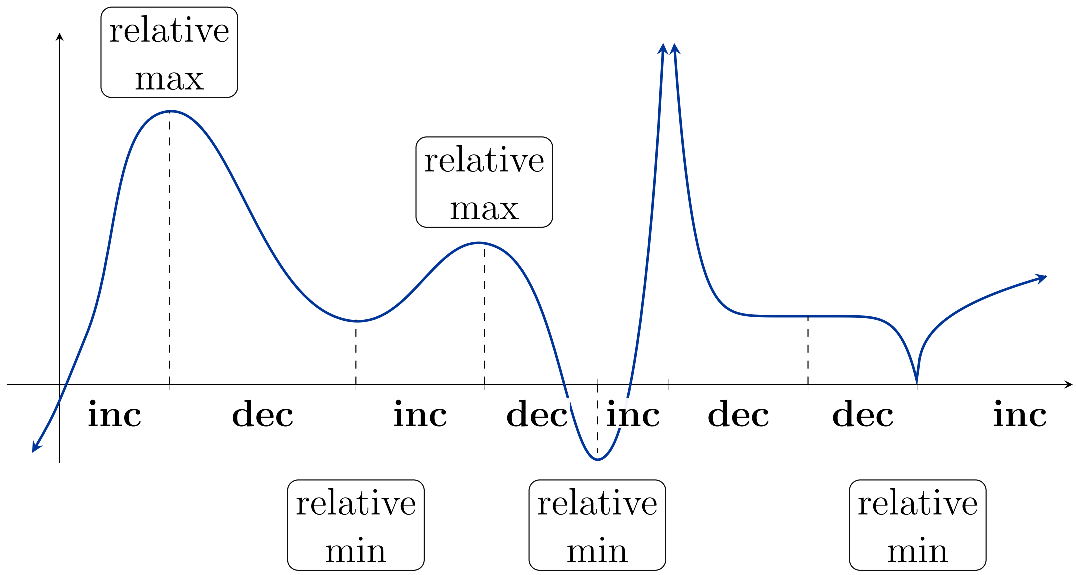
Definition.
Consider the function $f(x)$ on the interval (a, b). Given *any* two numbers x_1 and x_2 in (a, b) where $x_1 < x_2$, we say f is

increasing if $f(x_1) < f(x_2)$ decreasing if $f(x_1) > f(x_2)$



Definition. (Relative/Local Extrema)
A function f has a

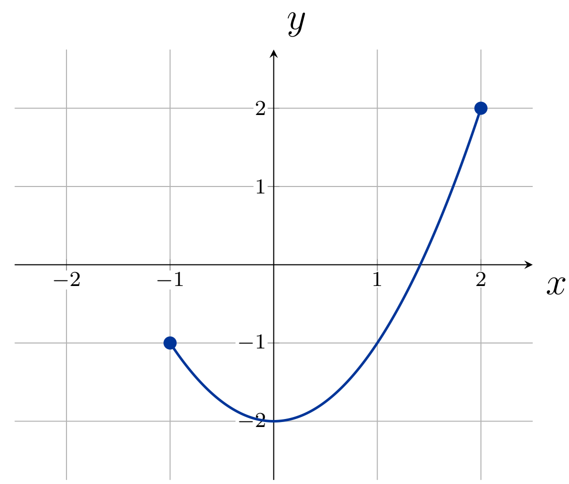
- **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in (a, b)
- **relative minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in (a, b)



Definition. (Absolute Extrema)
A function f defined on a domain D has an

- **absolute maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in D
- **absolute minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in D

Example. Using the following graph, find any relative and absolute extrema.



Example. Using the following graph, find any relative and absolute extrema.